Algebra 1 Name____________________
Unit 9 Date____________________
Lesson 8 Practice Period____________________

Alejandro's work when solving the equation $y = ax^2 - 6x + 9$ for a is shown. Use Alejandro's work to answer questions 1 and 2.

$$y = ax^2 - 6x + 9$$

$$y - (-6x + 9) = ax^2 - 6x + 9 - (-6x + 9)$$

$$y + 6x + 9 = ax^2$$

$$\frac{y + 6x + 9}{x^2} = \frac{ax^2}{x^2}$$

$$\frac{y + 6x + 9}{x^2} = a$$

1. Circle the step where Alejandro made an error.

2. Correct Alejandro's error and the steps that follow to show how to correctly isolate x in the equation. Use the space provided above to show the correct steps.

Solve each equation for the designated variable. Show your work.

3. Solve $-3(x + 5)(x + m) = y$ for m.	4. Solve $y = 2x^2 + bx - 9$ for b.

5. Solve $y = \frac{8}{3}(x+2)^2 + r$ for r.	6. Solve $y = -3(x-2)^2 + 4$ for x.
7. Solve $y = ax^2 + 7x - 9$ for a.	8. Solve $y = k(x-3)^2 - 7$ for k.
9. Solve $y = x^2 - 12x + 4w$ for w.	10. Solve $y = a(5x-1)(x+1)$ for a.